Defining causal effects

STSCI / INFO / ILRST 3900: Causal Inference

27 Aug 2026

Learning goals for today

By the end of class, you will be able to

- explain the fundamental problem of causal inference and the need for causal arguments
- define potential outcomes

Causal claims hinge on assumptions, not just data

[1]

[1] Photo credit: YantsImages via Wikipedia

Causal claims hinge on arguments, not just data

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   - Wearing two skates in Milan causes a person to win a gold medal.

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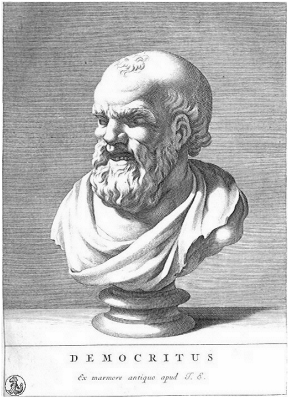
What do we mean when we say “cause”?

Deep philosophical question



I would rather discover one true cause than gain the kingdom of Persia.

Democritus (460-370bc)

Causal claims hinge on arguments, not just data

	Do you win gold if you:		Causal effect
	Wear skates	Do not wear skates	of wearing skates
Alysa Liu	Yes (1)	?	?
Sam	?	No (0)	?

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Observing

Image source: Wikimedia

Observation:

People who eat a Mediterranean diet have lower rates of cardiovascular disease

Observing

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Observation:

People who eat a Mediterranean diet have lower rates of cardiovascular disease

Discuss:

Do you think there is a causal effect here? If not, how might you explain away this apparent observation?

Fundamental problem of causal inference

Holland 1986

Descriptive evidence

average cardio status − average cardio status

	Outcome under Mediterranean diet	Outcome under prudent diet
Person 1	cardio status	
Person 2		cardio status
Person 3	cardio status	
Person 4		cardio status
Person 5	cardio status	
Person 6	cardio status	
Person 7		cardio status
Person 8	cardio status	

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Causal claim

average lifespan − average lifespan

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Descriptive evidence

average cardio status − average cardio status

	Outcome under Mediterranean diet	Outcome under prudent diet
Person 1	cardio status	missing
Person 2	missing	cardio status
Person 3	cardio status	missing
Person 4	missing	cardio status
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Person 6	cardio status	missing
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Causal claim

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Causal inference is a **missing data** problem

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Mathematical notation: Potential outcomes[2]

[2] Capital letters and lowercase letters mean different things!

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Y_i (observed/realized) Outcome Whether person i has cardio disease

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Mathematical notation: Potential outcomes[2]

Y_i	(observed/realized) Outcome	Whether person i has cardio disease
A_i	Treatment	Whether person i ate a Mediterranean diet

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Mathematical notation: Potential outcomes[2]

Y_i	(observed/realized) Outcome	Whether person i has cardio disease
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Y_i^a	Potential Outcome	Outcome person i would realize if **assigned** to treatment value a

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Examples:

If assigned prudent diet	If assigned mediterranean diet
Disease	No Disease

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$A_{\text{Sam}} = \texttt{MedDiet}$ We observed that Sam ate a Mediterranean diet

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$Y_{\text{Sam}} = \texttt{survived}$ We observe that Sam has no disease

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Y_i	(observed/realized) Outcome	Whether person i has cardio disease
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Examples:

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$Y_{\text{Sam}} = \texttt{survived}$ We observe that Sam has no disease

$Y_{\text{Sam}}^{\text{MedDiet}} = \texttt{survived}$ If Sam had been assigned a Mediterranean diet he would have no disease

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Y_i	(observed/realized) Outcome	Whether person i has cardio disease
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Disease	No Disease

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$Y_{\text{Sam}} = \texttt{survived}$	We observe that Sam has no disease
$Y_{\text{Sam}}^{\text{MedDiet}} = \texttt{survived}$	If Sam had been assigned a Mediterranean diet he would have no disease
$Y_{\text{Sam}}^{\text{PruDiet}} = \texttt{died}$	If Sam had been assigned a prudent diet he would have disease

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Practice

Using the slip of paper you received and the diet you follow, what is

- Y_i
- A_i
- Y_i^{MedDiet}
- Y_i^{PruDiet}

The consistency assumption

- **Consistency Assumption:** the observed outcome for an individual corresponds to the potential outcome for that individual's observed treatment

$$\underbrace{Y_i}_{\text{Observed Outcome}} = \underbrace{Y_i^{A_i}}_{\text{Potential Outcome for observed treatment}}$$

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- Consistency implies that observed outcome for an individual who "chose" a specific treatment is same as what we would have observed if the individual was "assigned" that treatment
- Consistency would not hold if "forcing" someone to eat a Mediterranean diet results in a different lifespan than what we observed for someone who chose the Mediterranean diet

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- Consistency would not hold if "forcing" someone to eat a Mediterranean diet results in a different lifespan than what we observed for someone who chose the Mediterranean diet
- Consistency implies treatments don't have hidden types
- Consistency would not hold if Italian olive oil resulted in a different lifespan than Greek olive oil

Practice: How would you say this in English?

We might wonder how a person's earnings relate to whether they hold a college degree

1. $\text{E}(\text{Earnings} \mid \text{Degree} = \texttt{TRUE}) > \text{E}(\text{Earnings} \mid \text{Degree} = \texttt{FALSE})$

2. $\text{E}(\text{Earnings}^{\text{Degree}=\texttt{TRUE}}) > \text{E}(\text{Earnings}^{\text{Degree}=\texttt{FALSE}})$

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   - Average earnings are higher among those with college degrees
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   - Average earnings are higher among those with college degrees
2. $\text{E}(\text{Earnings}^{\text{Degree}=\texttt{TRUE}}) > \text{E}(\text{Earnings}^{\text{Degree}=\texttt{FALSE}})$
   - The average earning if everyone was assigned to get a degree is higher than the average earnings if everyone was assigned to not get a degree
   - On average, getting a degree causes higher earnings

Practice:

1. On average, students who do the homework learn more than those who don't

2. On average, doing the homework causes more learning

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1. On average, students who do the homework learn more than those who don't

$$\text{E}(\text{Learning} \mid \text{HW} = \texttt{TRUE}) > \text{E}(\text{Learning} \mid \text{HW} = \texttt{FALSE})$$

2. On average, doing the homework causes more learning

$$\text{E}(\text{Learning}^{\text{HW}=\texttt{TRUE}}) > \text{E}(\text{Learning}^{\text{HW}=\texttt{FALSE}})$$

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You can now

- Read Chapter 1 of Hernán and Robins 2020